Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] **[18BS5AI01]**

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER – 2022

STATISTICS USING R

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.
- **iii.** Statistical tables should be provided.

1. a. Explain the construction of various Bar charts and Pie Charts. **07**

b. Construct the Histogram and give analysis for the following distribution. **07**

Marks	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60	60 - 70	70 - 80	80 - 90	90 - 100
Students	8	12	22	35	40	60	52	40	30	5

2. a. With the help of the given data **07**

Age (Years)	10 -14	15 -19	20 -24	25 -29	30 -34	35 -29
No. of persons	5	10	15	20	10	5

Find the following

- **i.** Values of middle 50% items.
- **ii.** Values of exactly 50% items.
- **iii.** Graphically using Ogive curve, the values of Q1, Q3 and median.

b. Determine Karl Pearson's coefficient of skewness for the following frequency distribution **07**

Variable	0-5	5-10	10-15	15-20	20-25	25-30	30-35	35-40
Frequency	2	5	7	13	21	16	8	3

And hence analyze the result.

3. a. By using given data obtain the weighted arithmetic mean index number and the weighted geometric mean index number. **07**

Commodity	Price in Rs./Unit		Weight
	Base year	Current year	
A	2	3	10
B	4	5	4
C	10	10	5
D	12	9	1

07

b. From the following data compute quantity indices by

i. Laspeyre' s method

ii. Paasche' s method

iii. Fisher' s method.

	2008		2010	
Commodity	Price	Total value	Price	Total value
A	10	100	12	180
B	12	240	15	450
C	15	225	17	340

OR

4. a. Compute Fisher's index number from the following data. Show that it satisfies both time reversal and factor reversal tests and discuss the results. **07**

Item	Base year		Current year	
	Price	Quantity	Price	Quantity
A	6.00	10	6.40	12
B	4.50	20	5.50	24
C	2.30	15	4.00	16
D	6.10	6	8.00	10

b. From the following data of whole sale prices of wheat for ten years construct index number taking a) 2000 as base b) by chain base method. **07**

Year	Price of wheat (Rs. Per 40 kg))	Year	Price of wheat (Rs. Per 40 kg))
2000	50	2005	78
2001	60	2006	82
2002	62	2007	84
2003	65	2008	88
2004	70	2009	90

5. a. Calculate the trend value to the following data by the method of semi-averages. **07**

Year	1991	1992	1993	1994	1995	1996
Sales (in Rs.1000)	60	75	81	110	106	120

b. Determine the seasonal variations by simple average method for the data given below. **07**

Quarter

Year	I	II	III	IV
2009	30	40	36	34
2010	34	52	50	44
2011	40	58	54	48
2012	54	76	68	62
2013	80	92	86	82

OR

6. a. Discuss irregular variations in context of time series. **05**

b. Fit a straight line trend by the method of least squares for the following data. **09**

Year	1983	1984	1985	1986	1987	1988
Sales (Rs. in lakhs)	3	8	7	9	11	14

Also estimate the sales for the year 1991

7. a. Describe a vector in R and different operations of vectors with examples. **04**

b. Write R commands, **05**

i. $A=\begin{bmatrix} 2 & 4 & 2 \\ 1 & 5 & 7 \end{bmatrix}$

ii. Extract element of 2nd row, 3rd column from matrix A.

iii. Extract elements of the entire 2nd row from matrix A.

iv. Transpose of matrix A and store in matrix B.

v. Combined matrix [3,7] with the matrix B.

c. Write R commands for the following **05**

i. The log to the base 10 of X1= 73.

ii. The square root of X2 = 137 by raising it to the power 0.5.

iii. For average of X1 and X2.

iv. If a= (8, 12, 5, 9) and b= (2, 4, 6, 8) are vectors. Write the command for C= $(5a+b)^2$.

OR

8. a. Using the data: **04**

12, 23, 25, 13, 45, 36, 58, 62.

Write R commands for the following

i. Mode

ii. Mean and Median of the given data which is less than 36.

b. Explain the basic data types in R with examples. **05**

c. Write the R commands for the following **05**

i. frequency distribution of the painter schools.

ii. bar graph of the painter schools and write a R command to color the bar chart.